

SHREYAS KARNAD

Ann Arbor, MI | +1 (734) 709-7325 | shreyas01karnad@outlook.com | linkedin.com/in/shreyas-karnad |
github.com/karnadsp | karnadsp.github.io

PROFESSIONAL SUMMARY

Health Informatics graduate of the University of Michigan (MHI, 3.936 GPA) and Biomedical Engineer with expertise in machine learning, generative AI pipelines, and healthcare data analytics. Built multi-agent LLM systems using CrewAI and DeepSeek, developed predictive models for clinical risk stratification, and designed FHIR-interoperable data workflows. Bridges technical depth in Python, R, and SQL with applied knowledge of clinical decision support, EHR systems, HIPAA compliance, and health data governance.

EDUCATION

University of Michigan – School of Information, Ann Arbor, MI Aug 2024 – May 2026

Master of Health Informatics (Machine Learning, Data Analytics) - GPA: 3.936

Focus: health data analytics, clinical decision support, FHIR interoperability, AI/ML in healthcare, data governance

Manipal Institute of Technology, MAHE, Manipal, India

Oct 2021 – Aug 2023

M.Tech, Biomedical Engineering (Biomedical Instrumentation) - GPA: 3.87

MIT School of Bioengineering Sciences & Research, MIT ADT, Pune, India

Aug 2017 – Aug 2021

B.Tech, Bioengineering (Biomedical Engineering elective) - GPA: 3.93

EXPERIENCE

University of Michigan – School of Public Health, Ann Arbor, MI

Jan 2026 – May 2026

Grader, Public Health Informatics (Part-time)

- Evaluated 30+ graduate-level assignments per cycle using structured rubrics, maintaining consistent scoring standards and reducing instructor review time through detailed written feedback on analytical approach and data interpretation. Collaborated directly with Prof. Rahul Ladhania on grading rubric design, dispute resolution, and quality assurance across multiple assignment cycles; identified and corrected systematic grading inconsistencies before grade release.

University of Michigan – School of Information, Ann Arbor, MI

Sep 2025 – Dec 2025

Instructional Aid, Health Informatics / Database Management (Part-time)

- Supported a 60+ student health informatics course by grading analytical examinations, proctoring assessments, and communicating complex technical concepts clearly to students with varying programming backgrounds.
- Collaborated directly with Prof. Bobby Madamanchi on exam paper design, dispute resolution, and quality assurance across 3 exam cycles; identified and corrected systematic grading inconsistencies before grade release.

Biocomplexity Institute, Indiana University, Bloomington, IN (Remote)

Jun 2025 – Aug 2025

Data Science Intern

- Designed schema-based JSON pipelines to automatically generate syntactically valid CompuCell3D biological simulation code from research paper abstracts; documented zero-shot LLM failure modes (version mismatches, hallucinated APIs, incorrect biological parameters) to guide iterative pipeline refinement.
- Translated natural-language biological modeling questions into structured computational parameters (adhesion, chemotaxis, diffusion) via an NL → ontology → code framework, enabling non-programmer researchers to generate simulation-ready outputs.
- Re-engineered a 9-agent CrewAI system (Scientist, Team Leader, Coder, Ontology Expert, Biological Modeling Expert, Version Checker, Coverage Verification, Packager, Interpreter) to replace OpenAI with DeepSeek Replicate via a custom Python wrapper in VS Code, resolving API environment conflicts and reducing inference costs while maintaining code generation accuracy.

Siamaf Healthcare Pvt. Ltd., Bengaluru, India (Remote)

Aug 2023 – Aug 2024

Associate Biomedical Engineer

- Evaluated deep learning classification models (ResNet50, GoogLeNet, DenseNet) in MATLAB Deep Learning Toolbox to classify harmonic signature data from magnetic nanoparticle-based intraoperative sentinel lymph node detection; achieved 96–100% accuracy distinguishing tumor-present from tumor-absent nodes in early-stage testing, with bacterial species identification validated as a secondary application.
- Integrated MATLAB with LabVIEW to interface with MafPro magnetic field probe hardware; deployed ML inference pipelines to AWS S3-connected cloud servers, enabling remote clinics without high-end NPU hardware to offload classification tasks to cloud infrastructure and receive results in real time.
- Designed a ruggedized field enclosure in Fusion 360 housing a Surface Pro with NPU, battery pack, and CAT8 magnetic detector port; implemented a locked-down Windows 11 environment restricting end-user access to MafPro software only, producing a deployment-ready clinical device prototype.

Siamaf Healthcare Pvt. Ltd., Bengaluru, India

Aug 2022 – Jul 2023

Graduate Research Intern

- Conducted structured primary research with 20+ surgical oncologists to identify unmet clinical workflow needs; synthesized

findings into formal requirements that directly shaped the product development roadmap.

- Built and validated a magnetic probe prototype for intraoperative sentinel lymph node detection; analyzed experimental sensor data using time-series methods and contributed findings to M.Tech thesis and internal technical reporting.

TATA Elxsi, St. Paul, MN (Remote)

Jan 2021 – Jun 2021

Project Intern

- Analyzed investment data across 50+ digital health companies totaling over \$12B; built weighted scoring models to identify high-growth sectors including AI/ML, cloud computing, and IoT, delivering sector rankings to support strategic investment decisions.
- Mapped Software as a Medical Device (SaMD) regulatory frameworks across FDA and IMDRF pathways, producing a risk classification reference guide used to inform product development planning.

PROJECTS

EHR AI Hallucination Mitigation Strategy

Mar 2026 – Apr 2026

- Identified that AI-generated hallucinations in EHR transcription tools create undetected clinical documentation errors; proposed a Zero-Trust RAG framework with retrieval-based validation checkpoints, reducing reliance on unverified LLM outputs and improving documentation accuracy and HIPAA compliance.

Ctrl Alt Drink - Mobile Health Intervention for Student Alcohol Moderation

Jan 2026 – Apr 2026

- Designed a behavioral health app targeting student alcohol moderation; incorporated real-time behavioral tracking, anonymous peer support, and micro-goal frameworks grounded in evidence-based behavior change principles to address a gap in accessible, stigma-free campus health tools.

Structural Health Disparities and Sepsis Risk Analysis

Sept 2025 – Dec 2025

- Analyzed a large administrative California county-level healthcare dataset to determine structural drivers of sepsis risk; built a Random Forest classification model achieving 80% accuracy and multivariate regression identifying insurance coverage as the primary predictor of postoperative outcomes.
- Authored full statistical methods and results sections for academic submission, including data quality validation, feature engineering, and county-level health disparity visualizations.

Python-Based Diabetes Risk Assessment App + FHIR Output

Oct 2024 – Nov 2024

- Built an end-to-end ML pipeline for diabetes risk prediction; generated structured FHIR-formatted outputs to enable interoperable data sharing across health systems, demonstrating real-world applicability of clinical decision support tools.

Diabetes Risk Analysis - SI 544, University of Michigan

Sep 2024 – Dec 2024

- Applied hypothesis testing (two-sample t-tests) and multivariate linear regression in R to a 768-patient Pima Indians diabetes dataset; identified blood glucose ($p < 2.2e-16$) and BMI as statistically significant primary predictors while surfacing zero-value missingness as a systematic data quality issue affecting model reliability.

Breast Cancer Tumor Network Analysis - SI 507, University of Michigan

Feb 2025 – Apr 2025

- Compared K-Means and Spectral Clustering algorithms on the Wisconsin Diagnostic breast cancer dataset (32 features, 569 patients); Spectral Clustering outperformed on accuracy and runtime, while network centrality analysis identified `concave_points_mean` as the most diagnostically influential tumor feature across degree, closeness, and betweenness metrics.

IVF Success Prediction - Binary Classification, Manipal Institute of Technology

April 2022 – June 2022

- Applied Logistic Regression, K-Nearest Neighbour, and Naïve Bayes classifiers in Python to a clinical IVF dataset (55 patients, 14 metabolites); KNN achieved best accuracy of 83% on optimized metabolite combinations; identified data volume as primary constraint on model reliability.

PPI2 Web Server - Protein-Protein Interaction Database

Jan 2019 – Dec 2020

- Developed an HTML-based research data system interfacing with a relational biological database; integrated interactive 3D protein interaction visualizations to support researcher queries and data exploration.

PAPERS

Current Trends of AI and Applications in Digital Pathology: A Comprehensive Review

Dec 2023

- *International Journal of Advanced and Applied Sciences (IJAAS)* - peer-reviewed publication

TECHNICAL SKILLS

Programming: Python (pandas, scikit-learn, matplotlib, seaborn), R (tidyverse, ggplot2, caret), SQL, MATLAB, LabVIEW | **GenAI/LLM:** CrewAI, OpenAI APIs, DeepSeek Replicate, prompt engineering, multi-agent workflows, RAG, ontology extraction | **ML/Stats:** predictive modeling, classification, regression, clustering, hypothesis testing, biostatistics, time-series analysis, harmonic signal analysis | **Deep Learning:** ResNet50, GoogLeNet, DenseNet, MATLAB Deep Learning Toolbox, image classification, biological signal classification | **Healthcare/Interop:** FHIR (file generation), HL7, HIE architecture, EHR systems, clinical decision support, HIPAA, 42 CFR | **Data:** MySQL, Microsoft Access, Excel; data pipeline design, EDA, feature engineering, data quality assurance | **Cloud/Tools:** AWS (S3, server-based inference), Git, VS Code, Fusion 360; technical documentation, reproducible workflows | **Languages:** English (fluent), Hindi/Hindustani (fluent), Marathi (native)